DEBAPRASAD PAUL

Senior Frontend Engineer | Real-Time & Data-Dense Web Systems
Bengaluru, India | Open to Remote (IST: full EU overlap, US-East to 12:00 PM ET) | +91 7005198655
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PROFESSIONAL SUMMARY

Senior Frontend Engineer with **5 years** of experience building real-time, data-dense web applications for banking, payments, and securities finance. Currently at **EquiLend**, where I own the frontend architecture of high-throughput securities lending platforms built on **React, TypeScript, WebSockets, and Module Federation microfrontends**. Specialized in low-latency rendering of streaming market data (virtualized grids, Core Web Vitals, P99 render latency), scalable design systems, and **LLM-streaming user interfaces**. Practitioner and internal champion of agentic AI development workflows (Claude Code, Cursor, MCP) as a measurable engineering-velocity multiplier.

TECHNICAL SKILLS

- **Languages & Core:** TypeScript, JavaScript (ES6+), HTML5, CSS3, SASS, Tailwind CSS, WebGL/GLSL
- **Frameworks & Architecture:** React 18/19, Next.js (App Router, React Server Components, Server Actions), Microfrontends, Webpack Module Federation, Monorepos (Turborepo/Nx), Design Systems, Component-Driven Architecture, Progressive Web Apps
- **Real-Time & Performance:** WebSockets, Server-Sent Events (SSE), LLM token streaming, List & grid virtualization (TanStack Virtual, react-window), Core Web Vitals (LCP, INP, CLS), Rendering & re-render profiling, Code splitting, Bundle optimization, Service Workers & caching, Lighthouse
- **State & Data Layer:** Redux, Redux Toolkit, Zustand, TanStack Query (React Query), Apollo Client, GraphQL, REST, Optimistic updates, Cache normalization, Offline-first persistence
- **AI Engineering:** Anthropic & OpenAI APIs, Streaming chat/agentic UIs, Model Context Protocol (MCP), Claude Code, Cursor, GitHub Copilot, Codex, LLM-assisted test generation
- **Testing & Quality:** Jest, Vitest, React Testing Library, Cypress, Playwright, Visual regression (Chromatic, Percy), Storybook, ESLint, Husky
- **Accessibility & Security:** WCAG 2.1 AA, ARIA, Screen-reader & keyboard-first UX, OAuth 2.0, JWT, SSO, RBAC
- **Tooling & Cloud:** Vite, Webpack, npm/yarn/pnpm, Git, GitHub Actions, Jenkins, AWS (S3, CloudFront, Lambda, EC2), NGINX, Sentry, Datadog RUM

EDUCATION

National Institute of Technology (NIT) Agartala — Agartala, India
B.Tech, Electronics and Communication Engineering — August 2017 – May 2021

PROFESSIONAL EXPERIENCE

EQUILEND | *Senior Software Development Engineer* — Bengaluru, India | 07/2025 – Present

- Reduced P99 grid-render latency from **200 ms to 120 ms** by implementing row virtualization and optimizing state updates under **sustained real-time data loads**.
- Decomposed a monolithic React application into **9 independently deployable microfrontends** using Webpack Module Federation, cutting release cycle time from **14 days to 7 days** and enabling **4 teams** to ship without a shared release train.
- Delivered **10+ reusable TypeScript components adopted across 9 applications**, reducing new-screen development time by **50%**.and eliminating UI drift across teams.
- Improved **WebSocket data reliability** by implementing reconnect handling and reconciliation logic to keep frontend state synchronized with incoming updates.
- Improved Core Web Vitals, **reducing LCP from 2.8s to 1.9s and INP from 280ms to 160ms** through route-level code splitting, deferred module loading, and render-path optimization, while **reducing JavaScript bundle size by 24%**.

- Drove adoption of **AI-assisted coding practices** within the frontend team, documenting usage guidelines and integrating them into existing development workflows. Built **custom MCP integrations to connect AI coding agents** with selected engineering tools and streamline context retrieval during development.
- Mentor frontend engineers through code reviews, design discussions, and collaborative debugging while promoting maintainable implementation patterns.

COMVIVA | *Senior Engineer* Bengaluru, India | 11/2023 – 06/2025

- Built scalable fintech UIs for business-critical modules — bank transfers, currency conversion, bulk payouts, and pooled-account systems with role-based access control — serving **130 million users across 80 clients**
- Integrated REST and GraphQL APIs using Apollo Client, improving frontend data access and consistency across fintech workflows.
- **Reduced redundant network round-trips by 60%** through client-side caching and API query optimization, improving data-fetching efficiency.
- Expanded automated frontend **test coverage to 85%** using Vitest, React Testing Library, and Jest, strengthening regression testing for critical workflows.
- Mentored **entry-level engineers** and instituted PR review standards, linting, and CI gates.

COMVIVA | *Product Development Engineer* Bengaluru, India | 07/2021 – 11/2023

- Developed a reusable component library in React, TypeScript, and modern JavaScript (ES6+), and optimized Redux-based state management — **reducing feature development effort by 35%** and **redundant API calls by 30%**.
- Engineered efficient API integration and data-handling strategies that **improved response times by 20%**, and hardened application stability through build optimization and CI/CD deployment via Jenkins and NGINX.
- Optimized API integration and Redux-based data handling to improve frontend responsiveness and reduce unnecessary requests.

PROJECTS

- **Samadhan — AI Civic Grievance Redressal Platform** (React, TypeScript, LLM streaming over SSE): Production-grade citizen complaint platform with AI-powered triage and routing, token-by-token streaming response UI, and a multi-theme accessible design system. **Live**
- **Atlantis — Fraud intelligence Mega platform UI** (React, Vite, Tailwind, IndexedDB) **Live**
- **CineOne — Discover films across time and space** (React, Three.js, OpenRouter, Claude Code): Browser-based film discovery platform with a deep-space aesthetic. **Live**
- **Billxo — Offline-First Invoicing PWA** (React, Redux, Service Workers): Progressive web app with offline-first data persistence, background sync, and installable mobile experience. **Live**
- **Logy.ai — AI Health Screening Platform** (Freelance; React, TypeScript, AWS): Delivered the complete Fullstack architecture for an AI-based health screening product from zero to production. **Live**
- **CheersPass — Event ticketing Platform** (Next.js, Stripe, Tailwind): A modern event ticketing platform that allows users to create, manage, and sell tickets for events. **Live**